
Many Doors, One Room: Independent Venues Converge

When Both Sides of a Creative Market Adapt

Syed Hamzah Rizvi
West Lafayette Jr/Sr High School
West Lafayette, IN 47906
syed105@purdue.edu

Abstract

The obvious remedy for a curator whose taste narrows a field is to add more curators. We test it. In an agent-based model, V venues with independently initialized tastes each accept work from a shared creator population and re-estimate their taste from their own accepted corpus, with total published volume held constant. Adding venues appears to help, until the confound is removed. Holding volume constant starves each venue of slots, and a venue estimating its taste from one or two accepted works produces a noisy estimate that stays scattered for reasons unrelated to editorial independence. Giving every venue a full corpus instead, ten independently initialized venues converge to a mean pairwise taste distance of 0.0001 and publish no more variety than a single gatekeeper: five matched venues are indistinguishable from one (0.421 against 0.427, $p = 0.21$) and ten are marginally lower (0.418, $p = 0.037$). The convergence is a dynamic rather than an artifact: venues begin by accepting nearly disjoint work (Jaccard overlap 0.156) and end accepting almost identical sets (0.995), and the effect survives a field more than three times larger in twice the dimensions. It requires both sides of the market to adapt. With creators frozen, venue divergence falls only from 0.970 to 0.494 and never collapses. Sweeping submitter-pool separation, only fully disjoint catchments prevent convergence; where the average creator submits to fewer than three of ten venues, divergence still falls 78%. Five venues that never re-estimate retain full divergence and publish 76% more variety than one ($p = 0.0001$). What matters is not how many doors exist but whether each door learns from what it has already admitted.

1 Introduction

A gatekeeper that learns its taste from the work it has already accepted trains on a record of its own decisions. In a companion simulation submitted concurrently, this narrows the published corpus beyond what selectivity alone produces: a self-referential gatekeeper reaches mean pairwise corpus distance 0.423 against 0.496 for a fixed gatekeeper of identical strictness ($p = 0.0001$), with both far below an open control at 0.804. That work names the single-gatekeeper assumption as its most consequential simplification, and this paper tests it.

The natural response to a curator with a drifting taste is structural. Give the field five curators, or ten, with different tastes and no obligation to agree. Plurality of venues is the mechanism by which real creative fields are supposed to avoid exactly this failure.

We find that it does not work, for a reason more interesting than the failure itself. Venues that begin with independent tastes and draw from a shared creator population converge on one another. They do so without communicating, without any shared objective, and without any of them behaving

unreasonably. Each simply follows its own corpus, while the population it draws from drifts toward whatever is being accepted.

The relevant prior literature treats monoculture as something decision-makers acquire by sharing. Kleinberg and Raghavan [2021] model screening under algorithmic monoculture, where many firms rely on the same ranking algorithm, and show that convergence on a single algorithm can reduce decision quality for the whole collection of agents even without shocks, and even when that algorithm is more accurate for any firm in isolation. Bommasani et al. [2022] formalize outcome homogenization and advance a component sharing hypothesis: systems built on the same data or models produce correlated outcomes for the same individuals, and they find empirically that increased data sharing reliably exacerbates homogenization. The literature contrasts this condition with algorithmic polyculture, in which decision-makers rely on independent systems.

Our venues are a polyculture by construction. They share no algorithm, no training data, and no communication channel; each is initialized independently and reads only its own corpus. The monoculture we observe is therefore not acquired but produced, arising in a system explicitly built to lack the component sharing that prior work identifies as the cause. What the venues share is only the population they draw from, and that turns out to be sufficient.

A second line of work concerns cultural markets. Salganik et al. [2006] show that social influence increases both inequality and unpredictability of outcomes, with quality only partly determining success. That result concerns which items win. Ours concerns whether the deciders remain distinguishable from one another.

Our contributions are three. First, evidence that a polyculture of independently initialized venues collapses endogenously into a monoculture under self-referential updating, verified against the obvious artifactual explanations, in a system that shares none of the components prior work identifies as the cause. Second, a methodological correction: the apparent benefit of adding venues under a fixed publication budget is estimation noise, and we nearly reported it as a finding. Third, an isolation of the mechanism showing that convergence requires adaptation on both the venue and creator side, so monoculture here is a coupled market failure rather than a property of gatekeepers, and that on the sharing side it requires only partial overlap of submitter pools.

2 Model

2.1 Scope of the operationalization

This is a simulation and the mapping onto real publishing, curation, or platform ecosystems is an argument rather than a measurement. Creators are points on a sphere rather than people; works are draws around those points; venues have a taste vector and nothing else, no reputation, budget, editorial board, or house style. The claim under test concerns the behavior of a mechanism under stated assumptions.

2.2 Setup

$N = 60$ creators occupy the unit sphere in $D = 8$ dimensions, positions c_i initialized uniformly at random. Each round, creator i proposes $w_i = \text{unit}(c_i + \sigma\epsilon)$ with $\sigma = 0.25$.

V venues hold taste vectors g_v , initialized independently and uniformly at random. Every creator submits to every venue. Venue v accepts its top s_v proposals by $w \cdot g_v$, accepted work enters that venue’s corpus (window 200), and the venue re-estimates g_v as the unit mean of its corpus.

Creators drift toward the centroid of everything accepted anywhere at rate $\mu = 0.06$, anchored to their own initial position at rate $\alpha = 0.10$:

$$c_i \leftarrow \text{unit}(c_i + \mu \text{unit}(\bar{a}) + \alpha c_{i,0}).$$

The anchor prevents the population collapsing to a point, which would make every condition identical; its calibration is discussed in the companion work.

2.3 Two ways to hold things constant, and why both are reported

Varying V while holding both total published volume and per-venue corpus size constant is impossible. Our first design held volume constant at $\text{TOTAL} = 15$ accepted works per round, so slots per venue fall as venues are added: 15 for $V = 1$, then 8 and 7, then 3 each, then five venues with 2 and five with 1 at $V = 10$ (the remainder is distributed to the lowest-indexed venues).

Under that design, adding venues appeared to help substantially. It does not. A venue with one or two slots estimates its taste from one or two works per round, and a noisy estimate stays scattered. Scatter across venues is what our divergence measure records, so estimation noise reads as editorial independence. We report this because it nearly became the paper’s headline.

The wide conditions give every venue the full 15 slots, matching per-venue estimation quality to $V = 1$ while letting total volume grow. Neither design is correct on its own. The contrast between them is where the result lives.

2.4 Measures

Union breadth. Mean pairwise cosine distance across all published work pooled from every venue: the field’s total variety.

Venue divergence. Mean pairwise cosine distance between venue taste vectors. Whether many deciders means many tastes.

Accepted-set overlap. Mean pairwise Jaccard overlap between the sets of proposals different venues accept in the same round. This is the diagnostic that separates a real dynamic from a construction artifact.

Confidence intervals are bootstrap percentile intervals over runs, runs as the independent unit; p -values are two-sided permutation tests on the difference in means [Efron and Tibshirani, 1993].

3 Results

3.1 Ten venues become one

Table 1: Main results, $n = 40$ runs per condition, 200 rounds. Brackets are bootstrap 95% CIs. The wide conditions match per-venue corpus size to $V = 1$; the others hold total volume constant.

Condition	Slots/venue	Union breadth	Divergence start	Divergence end
$V = 1$	15	0.427 [0.422, 0.433]	—	—
$V = 2$	8/7	0.360 [0.348, 0.375]	1.011	0.059 [0.032, 0.095]
$V = 5$	3	0.463 [0.443, 0.483]	0.977	0.317 [0.288, 0.346]
$V = 10$	2/1	0.648 [0.628, 0.667]	0.997	0.586 [0.558, 0.612]
$V = 5$ wide	15	0.421 [0.415, 0.428]	0.940	0.00022
$V = 10$ wide	15	0.418 [0.413, 0.424]	0.966	0.00012
$V = 5$ no update	3	0.750 [0.736, 0.764]	1.004	1.004 [0.976, 1.033]

With per-venue corpus size matched (Table 1, Figure 1B), ten independently initialized venues converge to a mean pairwise taste distance of 0.00012. They are, at the end of a run, numerically the same venue. Neither five nor ten such venues publish more variety than a single gatekeeper. Five matched venues reach union breadth 0.421 (95% CI 0.415–0.428) against 0.427 (0.422–0.433) for $V = 1$, difference -0.006 , $p = 0.21$, a null. Ten reach 0.418, difference -0.009 , $p = 0.037$, which is nominally significant in the wrong direction for plurality, though the effect is small on a scale where the no-update control sits at 0.750. The defensible summary is that matched-corpus plurality buys nothing, not that it costs something.

3.2 The correction: apparent plurality benefits are estimation noise

Under the volume-constant design (Table 1, upper rows), $V = 10$ reaches union breadth 0.648 against 0.427 for $V = 1$, a difference of $+0.221$ ($p = 0.0001$) that reads as a strong argument for plurality.

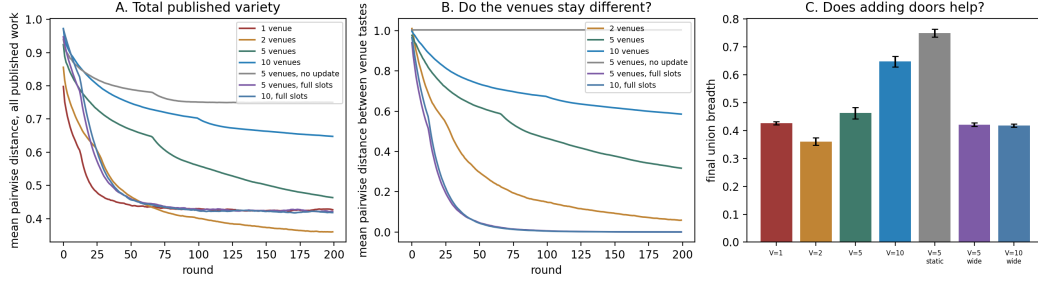


Figure 1: Total published variety over 200 rounds (A), divergence between venue tastes (B), and final union breadth by condition (C). $n = 40$ runs each. Panel B is the paper: every adaptive condition falls, and the matched-slot conditions fall to numerical identity, while venues that never re-estimate hold their initial divergence flat.

Matching per-venue corpus size removes it entirely: $V = 10$ wide against $V = 10$ gives -0.229 ($p = 0.0001$), landing back at the single-gatekeeper level.

The mechanism is straightforward once stated. Ten venues sharing a budget of fifteen slots get one or two each. A taste estimated from one or two works is mostly noise, and independent noise does not point in a common direction, so the venues stay scattered and their pooled output stays varied. That variety is sampling error wearing the costume of editorial diversity.

We report this at length because the artifact is more plausible than the finding and because the same confound is available to any empirical study comparing fields with different numbers of venues under a fixed total output.

3.3 Convergence is a dynamic, not a construction artifact

Every venue draws from the same 60 proposals, so it is reasonable to suspect their accepted sets were never distinct and identical corpora were forced from the start.

They were distinct. Mean pairwise Jaccard overlap between venues' accepted sets begins at 0.156 and ends at 0.995. In a wider field ($N = 200$ creators, $D = 16$) overlap begins at 0.042 and still climbs to 0.933, with divergence reaching 0.0008. Venues initially accept nearly disjoint work and end up accepting nearly the same work. Divergence crosses 1% of its initial value at round 85, gradually, so the collapse is not an initialization effect either.

3.4 Both sides of the market have to adapt

Freezing the creator population ($\mu = 0$) leaves venue divergence at 0.494 after 200 rounds, down from 0.970 but nowhere near collapse, with accepted-set overlap reaching only 0.511.

Venue self-reference alone is therefore not sufficient. Full convergence requires venues chasing their own corpora *and* creators chasing what gets accepted. Each loop alone leaves substantial editorial diversity intact; run together they eliminate it. Monoculture in this model is a coupled failure of the market rather than a property of gatekeepers, which is a different diagnosis from the one we set out to confirm.

3.5 Only complete separation prevents it

Every venue above draws from the same 60 creators, which is the model's least realistic feature. Real fields have catchments: submitters self-select, and a writer who sends to one journal may never send to another. We sweep this directly. Each creator is assigned a home venue and submits to each other venue with probability ω , so $\omega = 0$ gives fully disjoint pools and $\omega = 1$ reproduces the main experiment. Creators drift toward work accepted at venues they submitted to, since you cannot imitate what you never see.

Table 2: Venue divergence under partial pool separation, $V = 10$ with matched slots, $n = 20$ runs per cell, 200 rounds. Brackets are bootstrap 95% CIs.

ω	Mean venues per creator	Divergence start	Divergence end
0.0	1.0	1.005	1.016 [1.002, 1.032]
0.2	2.8	0.809	0.180 [0.164, 0.198]
0.4	4.6	0.883	0.076 [0.065, 0.088]
0.6	6.4	0.921	0.048 [0.039, 0.061]
0.8	8.2	0.935	0.019 [0.016, 0.023]
1.0	10.0	0.968	0.0002 [0.0000, 0.0003]

Fully disjoint pools prevent convergence completely (Table 2). Divergence at $\omega = 0$ ends at 1.016, marginally above where it started, so ten venues with private catchments remain ten distinct venues indefinitely.

Every other level of separation fails. At $\omega = 0.2$, where the average creator submits to fewer than three of ten venues, divergence still falls by 78% over the run. Partial separation buys a higher floor and does not change the direction of travel.

The reading we take is that the shared-pool assumption is not the fragile part of the main result. It is close to the weakest sufficient condition: convergence needs only that submitters overlap somewhat, which every real field satisfies. A field would have to be genuinely partitioned, with no creator ever submitting to more than one venue, to retain editorial diversity by this route.

3.6 What does preserve breadth

Five venues that never re-estimate their tastes retain their full initial divergence (1.004 at round 0 and at round 200; flat line in Figure 1B) and publish union breadth 0.750 against 0.463 for five adaptive venues, a difference of +0.287 ($p = 0.0001$), and 76% above a single adaptive gatekeeper.

The variable that matters is not the number of venues. It is whether a venue’s criteria are downstream of its own past acceptances.

4 Discussion

The structural intuition is that concentration is the problem and plurality the remedy. These results suggest the intuition is looking at the wrong variable. Ten venues that each learn from their own back catalogue, drawing on a population that is itself drifting toward whatever gets published, end up functionally identical. Nothing coordinated them. Nobody agreed to anything.

Read against the monoculture literature, the implication is that polyculture may not be a stable state. Prior work locates homogenization in shared components, a common algorithm [Kleinberg and Raghavan, 2021] or common data and models [Bommasani et al., 2022], which suggests that decision-makers running independent systems avoid the failure. Our venues run independent systems, share nothing, and converge anyway. Sharing an applicant pool was enough. If that holds outside simulation, auditing a field for component sharing would not detect this, and a field could satisfy every structural test for independence while its deciders became indistinguishable. Section 3.5 narrows what sharing is required: partial submitter overlap suffices, and only a completely partitioned field escapes.

The finding that both sides must adapt is the one we would build on. It says the locus of intervention is not necessarily the institution. A field whose creators hold their positions regardless of what is being published retains most of its editorial diversity even with self-referential venues, which relocates the question from institutional design toward whatever sustains independent creative commitment under commercial pressure.

We are conscious that our one clean intervention, venues that never update, is not a recommendation anyone could act on. No functioning venue can refuse to learn. The useful reading is comparative rather than prescriptive: it identifies which property carries the effect, and the design question it leaves open is what realistic mechanism has the same character. Candidates worth modeling include

venues that retain a founding charter with weight against their drifting corpus, rotating editorial staff whose tenure is shorter than the convergence timescale of roughly 85 rounds, and partly disjoint submitter pools.

5 Limitations

Simulation. No empirical corpus was analyzed. **Catchment structure.** Pool separation is swept in §3.5, but the assignment there is random rather than sorted: real catchments correlate with taste, since submitters choose venues they expect to suit them, and that correlation is untested and could cut either way. **Simultaneous submission.** Sequential submission with rejection carried forward is untested and is how real submission works. **The static control is not deployable.** It isolates a property rather than proposing a policy. **Inherited assumptions.** The creator update, anchor rate, and acceptance mechanics are carried from companion work and inherit its calibration history. **Unswep structure.** Corpus window, proposal noise, and acceptance fraction were held fixed; V and slot allocation were varied. **Uneven slots.** Under the volume-constant design the fixed budget does not divide evenly, so venues within a condition receive different allocations (8 and 7 at $V = 2$; five venues with 2 and five with 1 at $V = 10$). Venues within those conditions therefore differ in estimation quality as well as in taste, which is a further reason the volume-constant divergence figures should not be read as measuring editorial independence.

References

- [1] Bommasani, R., Creel, K.A., Kumar, A., Jurafsky, D. & Liang, P. (2022) Picking on the same person: does algorithmic monoculture lead to outcome homogenization? In *Advances in Neural Information Processing Systems*, vol. 35, pp. 3663–3678.
- [2] Efron, B. & Tibshirani, R.J. (1993) *An Introduction to the Bootstrap*. New York: Chapman & Hall.
- [3] Kleinberg, J. & Raghavan, M. (2021) Algorithmic monoculture and social welfare. *Proceedings of the National Academy of Sciences* **118**(22):e2018340118.
- [4] Salganik, M.J., Dodds, P.S. & Watts, D.J. (2006) Experimental study of inequality and unpredictability in an artificial cultural market. *Science* **311**(5762):854–856.

NeurIPS Paper Checklist

1. **Claims**
Question: Do the main claims accurately reflect the paper’s contributions and scope?
Answer: [Yes]
Justification: The abstract states the convergence result (0.00012 divergence), both matched-corpus contrasts against a single gatekeeper including the one that reaches nominal significance in the direction unfavourable to plurality ($p = 0.037$), the estimation-noise correction, and the two-sided-adaptation requirement. The correction to our own earlier reading is stated in the abstract rather than buried.
2. **Limitations**
Question: Does the paper discuss the limitations of the work?
Answer: [Yes]
Justification: Section 6 covers the simulation-to-reality gap, the shared submitter pool, simultaneous submission, the non-deployability of the static control, assumptions inherited from companion work, and unswept parameters.
3. **Theory assumptions and proofs**
Answer: [N/A]
Justification: The paper contains no theorems.
4. **Experimental result reproducibility**
Answer: [Yes]
Justification: Section 2 gives every parameter, both slot-allocation schemes, and all seven conditions. Seeds derive from loop indices and the simulation is deterministic given them, verified by repeated identical runs.
5. **Open access to data and code**
Answer: [Yes]
Justification: The simulation, the sensitivity diagnostics, the submitter-pool sweep, per-run result files, and the figure-generating code are released at <https://github.com/AIscend-Research/many-doors>. The code is three self-contained Python files depending only on NumPy and Matplotlib. Seeds are derived from loop indices and the reported numbers reproduce exactly.
6. **Experimental setting/details**
Answer: [Yes]
Justification: Sections 2.2–2.4 specify the full configuration and both diagnostic field sizes.
7. **Experiment statistical significance**
Answer: [Yes]
Justification: Every mean carries a bootstrap percentile 95% CI over runs ($n = 40$, runs as the independent unit); comparisons use two-sided permutation tests with exact p -values. The $V = 5$ wide against $V = 1$ null is reported as a null with its p -value.
8. **Experiments compute resources**
Answer: [Yes]
Justification: CPU-only agent-based simulation, no model training. The full set of reported runs plus diagnostics completes in a few minutes on a laptop.
9. **Code of ethics**
Answer: [Yes]
Justification: No personal data, no human subjects, no scraped material, no dual-use risk.
10. **Broader impacts**
Answer: [Yes]
Justification: Section 5 argues that structural remedies based on venue count may not address editorial convergence, and that studies comparing fields with differing venue counts under fixed output are vulnerable to the estimation-noise confound documented in Section 3.2. No artifact with foreseeable negative use is released.
11. **Safeguards**
Answer: [N/A]
Justification: No models or datasets are released.

12. **Licenses for existing assets**

Answer: [N/A]

Justification: NumPy and Matplotlib only, under their open licenses.

13. **New assets**

Answer: [N/A]

Justification: See item 5.

14. **Crowdsourcing and research with human subjects**

Answer: [N/A]

Justification: All agents are simulated.

15. **Institutional review board (IRB) approvals**

Answer: [N/A]

Justification: No human subjects research.

16. **Declaration of LLM usage**

Answer: [Yes]

Justification: No LLM is a component of the method; the study is agent-based simulation over synthetic vectors. Declaring anyway for transparency: an LLM assistant was used substantially during implementation and drafting, including writing simulation code, running experiments, and drafting prose. The research questions, the decision to add the matched-slot control after the initial confound was identified, and all interpretation are the author's. All reported numbers were produced by the released code rather than generated by a language model, and each numeric claim in this paper was checked against the released result files.